ALPHABET POSITION AND WORD MANIPULATION

Positions, Next-Letter Shifts and Vowels

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 29

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: alphabet position (English 2019 Q52) and next-letter vowel questions (Mathematics 2024 Q71 and Q98, Science 2024 Q77) appear.

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Four questions appear (English 2019 Q52, Mathematics 2024 Q71, Q98 and Science 2024 Q77). All four are solved below. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 4 PYQs and their types
2	Foundation	Alphabet table, left/right counting, shift-and-vowel trick
3	Solved PYQs	4 PYQs with step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Four questions, three papers

Paper	Q.	Question	Type	Ans
English 2019	52	What would be the 14th letter to the left of the twenty-first letter?	Alphabet position	B (G)
Mathematics 2024	71	JET LAG RUM PIN: change each letter to the next; how many words have no vowel?	Next-letter shift	B (Two)
Mathematics 2024	98	WON SEW HAT CUB: same question	Next-letter shift	C (Two)
Science 2024	77	TRIP SPIN PINK TASK: same question	Next-letter shift	3 (Zero)
English 2022 / Social Studies 2023	–	none		

PYQ INSIGHT: Three of the four PYQs are the **same question type**: change every letter to the next letter and count the words with no vowel. There is a **one-second trick**: only **Z, D, H, N, T** become vowels, so a word has no vowel after the shift if it contains none of these letters.

Pattern	PYQs	Priority
Next-letter shift and vowels in the new word	3	★★★ Must
Position of a letter to the left or right of another	1	★★★ Must
Reversing positions, opposite letters	0	★★ Expected
Rearranging a word alphabetically	0	★★ Expected

CHAPTER 2

Foundation

Alphabet table and the shift trick

2.1 Alphabet positions

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

Rule	Formula
Position from the right end	27 − position from the left
Opposite letter (same distance from the other end)	Letter at 27 − position (A–Z, B–Y, C–X ...)
n-th letter to the right of a letter at position p	Position p + n
n-th letter to the left of a letter at position p	Position p − n

NOTE: Vowels are at positions 1, 5, 9, 15 and 21 (A, E, I, O, U). Do not count the starting letter when counting “to the left” or “to the right”.

2.2 Next-letter shift and vowels

After the shift...	Which original letters give a vowel
Each letter changes to the next (A to B, ... Z to A)	Z, D, H, N, T (because A, E, I, O, U are the next letters)
Each letter changes to the previous	B, F, J, P, V (their previous letters are vowels)

SHORTCUT: JET: T becomes U (vowel). LAG: no D, H, N, T, Z: no vowel. RUM: none: no vowel. PIN: N becomes O (vowel). So two words have no vowel (Maths 2024 Q71).

CHAPTER 3

Solved PYQs

All four questions with step-by-step reasoning

English 2019 Q52

What would be the 14th letter to the left of the twenty-first letter of the English alphabet? (A) F (B) G (C) E (D) H

1. The 21st letter is U.
2. Counting to the left means going back: position 21 − 14 = 7.
3. The 7th letter is G.
Answer: Option B (G)

Mathematics 2024 Q71

Words: JET LAG RUM PIN. If each letter is changed to the next letter in the English alphabetical order, how many words thus formed will have no vowel? (A) Four (B) Two (C) One (D) Three

1. JET becomes KFU (U is a vowel): has a vowel.
2. LAG becomes MBH: no vowel. RUM becomes SVN: no vowel.
3. PIN becomes QJO (O is a vowel): has a vowel.
4. Two words (MBH and SVN) have no vowel.
Answer: Option B (Two)

Mathematics 2024 Q98

Words: WON SEW HAT CUB. Each letter is changed to the next letter. How many words thus formed will have no vowel? (A) One (B) Four (C) Two (D) Three

1. WON becomes XPO (O is a vowel).
2. SEW becomes TFX: no vowel.
3. HAT becomes IBU (I and U are vowels).
4. CUB becomes DVC: no vowel.
5. Two words have no vowel.
Answer: Option C (Two)

Science 2024 Q77

Words: TRIP SPIN PINK TASK. If each letter is changed to the next letter, how many words thus formed will have no vowel? (1) Three (2) Two (3) Zero (4) One

1. TRIP becomes USJQ (U is a vowel).
2. SPIN becomes TQJO (O is a vowel).
3. PINK becomes QJOL (O is a vowel).
4. TASK becomes UBTL (U is a vowel).
5. Every new word has a vowel, so the number of words with no vowel is zero. (Shortcut: each word contains T or N.)
Answer: Option 3 (Zero)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Next-letter vowels:** only Z, D, H, N, T change into vowels.
- **Previous-letter vowels:** only B, F, J, P, V.
- **Left/right counting:** add or subtract, do not count the starting letter.
- **Opposite letters add up to 27.**
- **Time:** 30 seconds per question.

Trap	How to avoid it
Forgetting Z becomes A	Z is followed by A, a vowel.
Counting the starting letter	14th letter to the left means position − 14.
Confusing the old and new vowels	The question asks about vowels in the **new** word.
Counting from the wrong end	Position from the right = 27 − position from the left.

Scope

Related PYQs / topics	Where
Series of letter clusters (shifts by a fixed step)	Series / Coding (earlier topics)
Letter-number-symbol series	Alphanumeric Series and Number Puzzles (Topic 25)

CHAPTER 5

Practice Set

6 original questions; answers on the next page

A. Next-letter shift

Q1. Each letter in each of the words MOLD FISH BALL KEPT is changed to the next letter in the English alphabetical order. How many words thus formed will have no vowel?

A. Zero B. One C. Two D. Three

Q2. Each letter in each of the words CRAB MILK FROG SEND is changed to the next letter in the English alphabetical order. How many words thus formed will have no vowel?

A. Zero B. One C. Two D. Three

B. Position questions

Q3. Which is the 9th letter to the right of the 12th letter from the left of the English alphabet?

A. U B. T C. V D. S

Q4. Which letter is the 8th from the right end of the English alphabet?

A. R B. T C. S D. Q

Q5. How many pairs of letters in the word STRANGE have as many letters between them in the word as in the English alphabet? (In either direction)

A. 0 B. 1 C. 2 D. 3

Q6. Which letter is the opposite of K (the same distance from the other end of the alphabet)?

A. L B. O C. Q D. P

Answer key

Q	Answer	Reason
1	B	Only the letters Z, D, H, N, T turn into a vowel (A, E, I, O, U) after the shift. Words containing none of these have no vowel after the shift: one (checked by program).
2	D	Only the letters Z, D, H, N, T turn into a vowel (A, E, I, O, U) after the shift. Words containing none of these have no vowel after the shift: three (checked by program).
3	A	The 12th letter is L. 12 + 9 = 21, the 21st letter is U.
4	C	Position from the left = 27 − 8 = 19, which is S.
5	B	Checked by program: pairs whose alphabet distance equals their distance in the word. Count = 1.
6	D	K is the 11th letter; 27 − 11 = 16, the 16th letter is P.

CHAPTER 6

Quick Revision

One page for the last day

Item	Rule
Position from the right	27 − position from the left
Next letter gives a vowel	Z, D, H, N, T
Previous letter gives a vowel	B, F, J, P, V
n-th to the right	Position + n
n-th to the left	Position − n
Vowels	A E I O U = 1, 5, 9, 15, 21

PYQ RULE: Three PYQs ask “next letter: how many words have no vowel”: use Z, D, H, N, T. Eng 2019 Q52: 21 − 14 = 7 = G.

ROUTINE: Exam-day routine: for shifts, scan each word for Z, D, H, N, T; for positions, use the position number and add or subtract. 30 seconds per question.